bathed in hot pools, where geothermally heated water reaches the surface, since Paleolithic times. Ancient Romans made use of it to heat their villas. Today it is used to generate electricity in more than 30 countries, with the United States, the Philippines, and Indonesia the world's leading producers.

Geothermal heat is also utilized directly to heat homes and roads in Iceland. Technology is now being developed that will use geothermal hot water to operate desalination plants. The only drawback of this renewable energy source is that it is concentrated near tectonic plate boundaries, where Earth's mantle heat rises close to the surface. The potential is much greater, but drilling for deep resources is very expensive.

"… the wind and the sun and the earth itself provide fuel that is free, in amounts that are effectively limitless."

Al Gore

American environmentalist and former US Vice President

Water power

Since water is 800 times denser than air, even a slow-moving flow can yield considerable amounts of energy if harnessed, for instance, by dams or tidal barrages that drive turbines connected to generators. China is the biggest producer of hydroelectric power (HEP), with 45,000 small installations in addition to "big-dam" schemes such as the Three Gorges project, whose 32 giant turbines have the capacity to produce 22,500 megawatts of electricity. A downside of big HEP schemes is that reservoirs created upstream of the dam can flood good farmland, forcing people to relocate and destroying ecosystems. Currently, 14 percent of the world's electricity comes from hydropower.

Tidal power is based on the same principle: moving water turns turbines, which drive electricity generators. The source of energy from a tidal scheme is reliable, generating power each time the tide ebbs and flows, but such